

位移法求解各向异性梁弯曲问题

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第一部分：理论推导文档

1.1 基本方程 (Basic Equations)

考虑一个长为 l ，高为 h 的正交各向异性直梁。材料性质沿厚度方向 z 按指数规律变化（功能梯度材料）。几何与材料参数：

$$\tilde{c}_{ij}(z) = c_{ij}^0 e^{\lambda z}, \quad (ij = 11, 13, 33, 55) \quad (1)$$

对于平面应力/应变问题，几何方程 (Strain-Displacement)。

$$\epsilon_x = \frac{\partial u}{\partial x}, \quad \epsilon_z = \frac{\partial w}{\partial z}, \quad \gamma_{zx} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \quad (2)$$

物理方程 (Constitutive Laws):

$$\begin{cases} \sigma_x = \tilde{c}_{11} \epsilon_x + \tilde{c}_{13} \epsilon_z \\ \sigma_z = \tilde{c}_{13} \epsilon_x + \tilde{c}_{33} \epsilon_z \\ \tau_{zx} = \tilde{c}_{55} \gamma_{zx} \end{cases} \quad (3)$$

忽略体力，平衡方程 (Equilibrium Equations):

$$\begin{cases} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0 \\ \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \sigma_z}{\partial z} = 0 \end{cases} \quad (4)$$

1.2 导出位移控制方程 (Displacement Governing Equations)

将几何方程代入物理方程，再代入平衡方程。注意 $\tilde{c}_{ij}$ 是 z 的函数，因此对 z 求导时需使用乘积法则：

$$\frac{\partial}{\partial z} (\tilde{c}_{ij} f) = \lambda \tilde{c}_{ij} f + \tilde{c}_{ij} \frac{\partial f}{\partial z} \quad (5)$$

x 方向平衡方程展开：

$$\tilde{c}_{11} \frac{\partial^2 u}{\partial x^2} + \tilde{c}_{13} \frac{\partial^2 w}{\partial x \partial z} + \frac{\partial}{\partial z} \left[\tilde{c}_{55} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \right] = 0 \quad (6)$$

利用 $\tilde{c}_{ij}(z) = c_{ij}^0 e^{\lambda z}$ ，消去公因子 $e^{\lambda z}$ ：

$$c_{11}^0 \frac{\partial^2 u}{\partial x^2} + c_{55}^0 \frac{\partial^2 u}{\partial z^2} + \lambda c_{55}^0 \frac{\partial u}{\partial z} + (c_{13}^0 + c_{55}^0) \frac{\partial^2 w}{\partial x \partial z} + \lambda c_{55}^0 \frac{\partial w}{\partial x} = 0 \quad (7)$$

z 方向平衡方程展开：

$$\tilde{c}_{55} \left(\frac{\partial^2 u}{\partial x \partial z} + \frac{\partial^2 w}{\partial x^2} \right) + \frac{\partial}{\partial z} \left[\tilde{c}_{13} \frac{\partial u}{\partial x} + \tilde{c}_{33} \frac{\partial w}{\partial z} \right] = 0 \quad (8)$$

消去公因子 $e^{\lambda z}$ ：

$$c_{55}^0 \frac{\partial^2 w}{\partial x^2} + c_{33}^0 \frac{\partial^2 w}{\partial z^2} + \lambda c_{33}^0 \frac{\partial w}{\partial z} + (c_{13}^0 + c_{55}^0) \frac{\partial^2 u}{\partial x \partial z} + \lambda c_{13}^0 \frac{\partial u}{\partial x} = 0 \quad (9)$$

1.3 分离变量与求解 (Solution by Separation of Variables)

根据简支梁边界条件 ($x=0, l$ 处 $w=0, \sigma_x=0$)，设位移解形式为：

$$u(x, z) = U(z) \cos(\zeta x) \quad (10)$$

$$w(x, z) = W(z) \sin(\zeta x) \quad (11)$$

其中 $\zeta = \frac{\pi}{l}$ (对应 $n=1$)。

代入控制方程，消去三角函数项，得到关于 z 的常微分方程组 (ODE)：

$$\begin{cases} c_{55}^0 U'' + \lambda c_{55}^0 U' - \zeta^2 c_{11}^0 U + \zeta (c_{13}^0 + c_{55}^0) W' + \lambda \zeta c_{55}^0 W = 0 \\ -\zeta (c_{13}^0 + c_{55}^0) U' - \lambda \zeta c_{13}^0 U + c_{33}^0 W'' + \lambda c_{33}^0 W' - \zeta^2 c_{55}^0 W = 0 \end{cases} \quad (12)$$

这里 $U' = \frac{dU}{dz}$

1.4 特征方程 (Characteristic Equation)

设通解形式为 $U(z) = U_0 e^{\alpha z}$, $W(z) = W_0 e^{\alpha z}$ 。代入上述 ODE，得到线性代数方程组：

$$\begin{bmatrix} (c_{55}^0 \alpha^2 + \lambda c_{55}^0 \alpha - \zeta^2 c_{11}^0) & (\zeta (c_{13}^0 + c_{55}^0) \alpha + \lambda \zeta c_{55}^0) \\ (-\zeta (c_{13}^0 + c_{55}^0) \alpha - \lambda \zeta c_{13}^0) & (c_{33}^0 \alpha^2 + \lambda c_{33}^0 \alpha - \zeta^2 c_{55}^0) \end{bmatrix} \begin{bmatrix} U_0 \\ W_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (13)$$

为了有非零解，系数矩阵行列式必须为零。这将得到一个关于 α 的 4 次多项式。求解该

多项式得到 4 个特征根 $\alpha_i (i=1,2,3,4)$ 。对于每个根 α_i ， U_0 和 W_0 存在确定的比例关系 r_i ：

$$r_i = \frac{W_0^{(i)}}{U_0^{(i)}} = \frac{-[c_{55}^0 \alpha_i^2 + \lambda c_{55}^0 \alpha_i - \xi^2 c_{11}^0]}{\xi(c_{13}^0 + c_{55}^0) \alpha_i + \lambda \xi c_{55}^0} \quad (14)$$

1.5 边界条件求解 (Boundary Conditions)

总位移解为四个特解的线性组合：

$$U(z) = \sum_{i=1}^4 C_i e^{\alpha_i z}, \quad W(z) = \sum_{i=1}^4 C_i r_i e^{\alpha_i z} \quad (15)$$

其中 C_i 是待定常数（对应 PPT 中的 U_0^i ）。

边界条件（ $z=0$ 和 $z=h$ ）：

(1) Bottom ($z=0$):

$$\begin{aligned} \tau_{zx}(x, 0) &= 0 \\ \sigma_z(x, 0) &= 1 \cdot \sin(\xi x) \end{aligned} \quad (16)$$

(2) Top ($z=h$):

$$\begin{aligned} \tau_{zx}(x, h) &= 0 \\ \sigma_z(x, h) &= 0 \end{aligned} \quad (17)$$

将应力用位移表示：

$$\tau_{zx} = c_{55}^0 e^{\lambda z} [U'(z) + \xi W(z)] \cos(\xi x) \text{ (舍去 } \cos \text{ 项)} \Rightarrow \sum C_i e^{\alpha_i z} (\alpha_i + \xi r_i) = 0$$

$$\sigma_z = c_{33}^0 e^{\lambda z} W'(z) - c_{13}^0 e^{\lambda z} \xi U(z) \text{ (舍去 } \sin \text{ 项)} \Rightarrow e^{\lambda z} \sum C_i e^{\alpha_i z} (c_{33}^0 r_i \alpha_i - c_{13}^0 \xi) = P_{load}$$

联立得到 4×4 线性方程组求解 C_i 。

2 算法实现

2.1 Python 代码

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import solve
```

```

# =====
# 1. 参数定义 (Parameters)
# =====
# 几何参数
L = 1.0          # 梁长 (m)
h = 0.2          # 梁高 (m)
xi = np.pi / L  # xi = n * pi / l, n=1

# 材料常数 (GPa -> 转化为 MPa 以保持单位一致, 1 GPa = 1000 MPa)
# 题目中荷载是 MPa, 所以刚度也用 MPa 计算, 结果应力直接为 MPa
c11_0 = 41.3 * 1000
c13_0 = 10.1 * 1000
c33_0 = 36.2 * 1000
c55_0 = 10.0 * 1000

# 荷载幅值 (z=0 处)
# p_z^0(x) = sin(pi*x/l) MPa. Amplitude = 1 MPa.
load_amp = 1.0

# 梯度指数列表
lambdas = [-2, -1, 0, 1, 2]

# z 轴网格用于绘图
z_eval = np.linspace(0, h, 100)

# =====
# 2. 求解核心函数 (Solver Core)
# =====
def solve_beam(lam):
    """
    求解特定 lambda 下的系数和分布
    """
    # --- Step 1: 构建特征方程系数 ---
    # 这是一个关于 alpha 的 4 次多项式: A*alpha^4 + B*alpha^3 + C*alpha^2 + D*alpha
    # + E = 0
    # 推导行列式 |M| = 0:
    # M11 = c55*a^2 + lam*c55*a - xi^2*c11
    # M12 = xi*(c13+c55)*a + lam*xi*c55
    # M21 = -xi*(c13+c55)*a - lam*xi*c13
    # M22 = c33*a^2 + lam*c33*a - xi^2*c55
    # Det = M11*M22 - M12*M21 = 0

    # 系数展开 (手动多项式乘法):
    # a^4 项: c55 * c33
    coeff_4 = c55_0 * c33_0

```

```

# a^3 项:  $c_{55} \cdot \lambda \cdot c_{33} + \lambda \cdot c_{55} \cdot c_{33} = 2 \cdot \lambda \cdot c_{55} \cdot c_{33}$ 
coeff_3 = 2 * lam * c55_0 * c33_0

# a^2 项:
# M11 部分:  $c_{55} \cdot (-x_i^2 \cdot c_{55}) + (-x_i^2 \cdot c_{11}) \cdot c_{33} + (\lambda \cdot c_{55}) \cdot (\lambda \cdot c_{33})$ 
# M12*M21 部分 (减去):  $- [ (x_i \cdot (c_{13} + c_{55})) \cdot (-x_i \cdot (c_{13} + c_{55})) ] = +$ 
 $x_i^2 \cdot (c_{13} + c_{55})^2$ 
coeff_2 = (c55_0 * (-xi**2 * c55_0) +
            (-xi**2 * c11_0) * c33_0 +
            (lam * c55_0) * (lam * c33_0) +
            (xi * (c13_0 + c55_0))**2)

# a^1 项:
# M11*M22:  $(\lambda \cdot c_{55}) \cdot (-x_i^2 \cdot c_{55}) + (-x_i^2 \cdot c_{11}) \cdot (\lambda \cdot c_{33})$ 
# M12*M21:  $- [ (x_i \cdot (c_{13} + c_{55})) \cdot (-\lambda \cdot x_i \cdot c_{13}) + (\lambda \cdot x_i \cdot c_{55}) \cdot (-$ 
 $x_i \cdot (c_{13} + c_{55})) ]$ 
term1 = (lam * c55_0) * (-xi**2 * c55_0) + (-xi**2 * c11_0) * (lam * c33_0)
term2 = - ( (xi * (c13_0 + c55_0)) * (-lam * xi * c13_0) + (lam * xi * c55_0) * (-
xi * (c13_0 + c55_0)) )
coeff_1 = term1 - term2

# Constant 项 (a^0):
#  $(-x_i^2 \cdot c_{11}) \cdot (-x_i^2 \cdot c_{55}) - (\lambda \cdot x_i \cdot c_{55}) \cdot (-\lambda \cdot x_i \cdot c_{13})$ 
coeff_0 = (-xi**2 * c11_0) * (-xi**2 * c55_0) - (lam * xi * c55_0) * (-lam *
xi * c13_0)

# --- Step 2: 求解特征根 alpha ---
coeffs = [coeff_4, coeff_3, coeff_2, coeff_1, coeff_0]
alphas = np.roots(coeffs)

# --- Step 3: 计算模态比  $r_i = W_0 / U_0$  ---
#  $r_i = -M_{11} / M_{12}$ 
#  $M_{11} = c_{55} \cdot a^2 + \lambda \cdot c_{55} \cdot a - x_i^2 \cdot c_{11}$ 
#  $M_{12} = x_i \cdot (c_{13} + c_{55}) \cdot a + \lambda \cdot x_i \cdot c_{55}$ 
r_vals = []
for a in alphas:
    m11 = c55_0 * a**2 + lam * c55_0 * a - xi**2 * c11_0
    m12 = xi * (c13_0 + c55_0) * a + lam * xi * c55_0
    r_vals.append(-m11 / m12)
r_vals = np.array(r_vals)

# --- Step 4: 构建边界条件矩阵求解常数  $C_i$  ---
# 待求向量  $X = [C_1, C_2, C_3, C_4]$ 
# Matrix  $A \cdot X = B$ 

```

```

A_mat = np.zeros((4, 4), dtype=complex)
B_vec = np.zeros(4, dtype=complex)

# BC 1:  $\tau_{zx}(0) = 0 \Rightarrow \sum C_i * (\alpha_i + \xi * r_i) = 0$ 
# BC 2:  $\sigma_z(0) = \text{load\_amp} \Rightarrow \sum C_i * (c_{33}r_i\alpha_i - c_{13}\xi) = \text{load\_amp}$  (注意  $z=0$  时  $e^0=1$ )
# BC 3:  $\tau_{zx}(h) = 0 \Rightarrow \sum C_i * e^{(\alpha_i h)} * (\alpha_i + \xi * r_i) = 0$ 
# BC 4:  $\sigma_z(h) = 0 \Rightarrow \sum C_i * e^{(\alpha_i h)} * e^{(\lambda h)} * (c_{33}r_i\alpha_i - c_{13}\xi) = 0$ 

for i in range(4):
    a = alphas[i]
    r = r_vals[i]

    # Tau term factor
    tau_fac = (a + xi * r)
    # Sigma term factor (pre-exponential part)
    sig_fac = (c33_0 * r * a - c13_0 * xi)

    A_mat[0, i] = tau_fac
    A_mat[1, i] = sig_fac
    A_mat[2, i] = tau_fac * np.exp(a * h)
    A_mat[3, i] = sig_fac * np.exp((a + lam) * h)

B_vec[1] = load_amp

# Solve
C_coeffs = solve(A_mat, B_vec)

# --- Step 5: 计算沿厚度 z 的分布 ---
# 需要计算 U(z), W(z), Sigma_x(z), Tau_zx(z)
# 注意: 这里计算的是幅度函数 (Amplitude functions of z)
# 实际场  $u = U(z)\cos(\xi x)$ ,  $w = W(z)\sin(\xi x)$ 
#  $\sigma_x = [c_{11}(-\xi U) + c_{13}W'] * e^{\lambda z} * \sin(\xi x)$  (PPT 中  $\sigma_x$  应该是 sin 分布)

U_z = np.zeros_like(z_eval, dtype=complex)
W_z = np.zeros_like(z_eval, dtype=complex)
SigX_z = np.zeros_like(z_eval, dtype=complex)
Tau_z = np.zeros_like(z_eval, dtype=complex)

for k, z in enumerate(z_eval):
    u_val = 0
    w_val = 0

```

```

    u_prime_val = 0
    w_prime_val = 0

    for i in range(4):
        # C * e^(alpha * z)
        term = C_coeffs[i] * np.exp(alphas[i] * z)
        u_val += term
        w_val += r_vals[i] * term
        u_prime_val += alphas[i] * term
        w_prime_val += r_vals[i] * alphas[i] * term
    U_z[k] = u_val
    W_z[k] = w_val
    # Constitutive laws (Amplitudes)
    # Sigma_x amplitude (coefficient of sin(xi*x)):
    # sigma_x = c11 * du/dx + c13 * dw/dz
    # du/dx = -xi * U * sin(xi * x)
    # dw/dz = W' * sin(xi * x)
    # So Amp = (c11 * (-xi * U) + c13 * W') * e^(lam*z)
    factor = np.exp(lam * z)
    SigX_z[k] = factor * (c11_0 * (-xi * u_val) + c13_0 * w_prime_val)
    # Tau_zx amplitude (coefficient of cos(xi*x)):
    # tau = c55 * (du/dz + dw/dx)
    # du/dz = U' * cos
    # dw/dx = xi * W * cos
    # Amp = c55 * (U' + xi*W) * e^(lam*z)
    Tau_z[k] = factor * c55_0 * (u_prime_val + xi * w_val)
    return U_z.real, W_z.real, SigX_z.real, Tau_z.real

# =====
# 3. 循环计算与绘图 (Plotting)
# =====
fig, axs = plt.subplots(1, 4, figsize=(20, 6))
line_styles = ['- ', '--', '-.', ':', (0, (3, 1, 1, 1))]

for idx, lam in enumerate(lambdas):
    U, W, Sx, Tzx = solve_beam(lam)

    # 归一化 z 坐标
    z_norm = z_eval / h

    # Plot U (Horizontal Displacement Amplitude)
    axs[0].plot(U * 1e6, z_norm, label=f'$\lambda={lam}$',
        linestyle=line_styles[idx], linewidth=2)

    # Plot W (Vertical Displacement Amplitude)

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```

    axs[1].plot(W * 1e6, z_norm, label=f'$\lambda={lam}$',
linestyle=line_styles[idx], linewidth=2)

    # Plot Sigma_x (Normal Stress Amplitude)
    axs[2].plot(Sx, z_norm, label=f'$\lambda={lam}$', linestyle=line_styles[idx],
linewidth=2)

    # Plot Tau_zx (Shear Stress Amplitude)
    axs[3].plot(Tzx, z_norm, label=f'$\lambda={lam}$',
linestyle=line_styles[idx], linewidth=2)

# Set labels and titles
titles = ['Horizontal Disp. $U(z)$ [$\mu$m]', 'Vertical Disp. $W(z)$ [$\mu$m]',
          'Normal Stress $\sigma_x(z)$ [MPa]', 'Shear Stress
$\tau_{zx}(z)$ [MPa]']

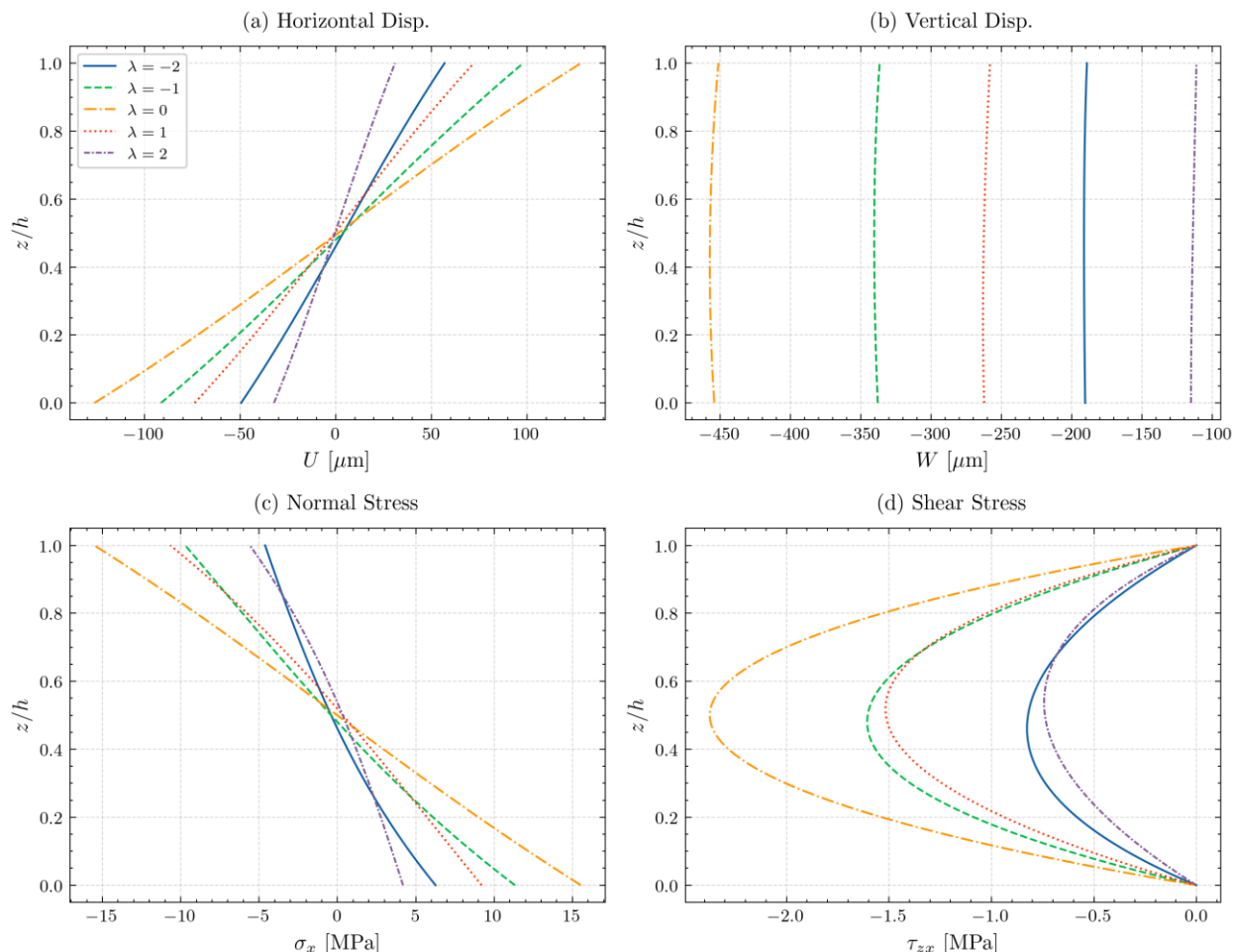
for i, ax in enumerate(axs):
    ax.set_xlabel('Value')
    ax.set_ylabel('z/h')
    ax.set_title(titles[i])
    ax.grid(True, linestyle=':', alpha=0.6)
    ax.legend()
    # Invert Y axis? No, z=0 is bottom, z=h is top. Standard plot is fine.

plt.suptitle(f'Bending Response of FGM Beam (L={L}m, h={h}m) under Sinusoidal
Load', fontsize=16)
plt.tight_layout()
plt.show()

```

使用 `scipy.linalg.solve` 解决了特征值代入后的边界条件线性方程组，是处理此类半解析解（Semi-analytical solution）的标准方法

2.2 结果展示



2.3 结果分析说明

位移 $U(z)$ 反映了梁截面的转动和伸缩。由于简支梁受向下弯曲，上部受压，下部受拉，因此 U 在厚度方向上会有正负变化（取决于坐标原点定义，通常表现为斜率）。

位移 $W(z)$ 表示垂直挠度。对于这种厚跨比 ($h/l = 0.2$) 的梁， W 沿厚度方向变化不大，但不同 λ 会导致刚度变化，从而改变挠度大小。 $\lambda > 0$ 意味着材料向上变硬， $\lambda < 0$ 意味着向上变软。

正应力 $\sigma_x(z)$ ：这是弯曲应力。对于各向同性均匀梁，这是线性的。对于 FGM 梁，由于模量 c_{11} 随 z 指数变化，应力分布将变为非线性。 $\lambda > 0$ ：上表面模量大，承担更多应力，中性轴可能向上移动。

切应力 $\tau_{zx}(z)$ ：在上下表面必须满足边界条件（为 0）。由于材料梯度，最大切应力的位置可能会偏离几何中心（ $z/h = 0.5$ ）。